Patent Proposal: Weight-Delta Vault Adapters (WDVA) for Privacy-Preserving, Per-User Small Language Model Personalization

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Patent Application Proposal
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Executive Summary

This document presents a novel invention for **Weight-Delta Vault Adapters (WDVA)**, a groundbreaking system for privacy-preserving personalization of small language models (SLMs) that enables secure, per-user model customization without compromising sensitive data privacy. The invention addresses critical limitations in current AI personalization approaches by introducing encrypted, user-scoped weight deltas combined with ephemeral runtime merging and consent-based access control.

Key Inventive Aspects:

- Encrypted per-user weight-delta vaults with ephemeral merging
- Model Context Protocol (MCP) gated consent management
- Genomics-aware uncertainty quantification with non-diagnostic safeguards
- Right-to-be-forgotten through cryptographic deletion
- Provenance ledger with end-to-end auditability

Technical Problem Statement

Current approaches to AI model personalization face fundamental limitations:

1. **Privacy Compromise**: Traditional fine-tuning comingles user data in shared model weights, making individual data extraction impossible and violating privacy principles.
2. **Revocation Impossibility**: Existing personalized models cannot effectively "forget" specific users without complete retraining, making GDPR Article 17 ("Right to be Forgotten") compliance technically infeasible.

3. **Cross-User Leakage**: Shared model architectures risk inadvertent disclosure of one user's sensitive information to another user through gradient-based attacks or inference techniques.
4. **Genomics Privacy Gap**: No existing system provides appropriate safeguards for genomic data integration in AI systems, particularly regarding uncertainty quantification and non-diagnostic boundaries.
5. **Consent Granularity**: Current systems lack fine-grained, purpose-limited, time-scoped consent mechanisms for AI model access to sensitive personal data.

Inventive Solution: WDVA Architecture

Core Technical Innovation

The Weight-Delta Vault Adapters (WDVA) system introduces a novel architecture comprising:

1. Encrypted Weight-Delta Vaults

Novel Aspect: Per-user weight deltas (ΔW_user) are generated through parameter-efficient fine-tuning (DoRA/LoRA) and immediately encrypted using authenticated encryption with associated data (AEAD) algorithms, specifically XChaCha20-Poly1305, before storage.

Technical Implementation:

- Weight deltas computed via: ΔW_user = W_personalized - W_base
- Encryption: E_k(ΔW_user || manifest || timestamp) → WDVA_blob
- Cryptographic binding to user identity and access policies

2. Ephemeral Runtime Merging

Novel Aspect: Personalized models are never persistently stored. Instead, base models are temporarily merged with decrypted weight deltas within a Trusted Execution Environment (TEE) for the duration of inference only.

Technical Process:

- Decrypt WDVA in volatile memory: D_k(WDVA_blob) → ΔW_user
- Ephemeral merge: W_temp = W_base ⊕ ΔW_user
- Inference execution with memory scrubbing post-completion
- Cryptographic zeroization of all temporary artifacts

3. MCP-Gated Consent Management

Novel Aspect: Integration with Model Context Protocol (MCP) to provide granular, cryptographically-enforced consent management with purpose and time limitations.

Technical Framework:

- JSON Web Token (JWT) based consent tokens with custom claims
- Hierarchical scope structures: `genomics:read:variant_analysis:30_days`
- Policy-based access control with runtime evaluation
- Revocable consent through token invalidation and key rotation

4. Genomics-Aware Processing Pipeline

Novel Aspect: Specialized handling of genomic data (VCF files) with uncertainty quantification and strict non-diagnostic boundaries.

Technical Components:

- Variant feature extraction with confidence intervals
- Population frequency analysis for uncertainty estimation
- Clinical significance filtering to prevent diagnostic claims
- Familial privacy protection through dependent differential privacy

5. Cryptographic Right-to-be-Forgotten

Novel Aspect: True data erasure through cryptographic key destruction rather than data deletion, providing immediate and verifiable forgetting.

Implementation Mechanism:

- WDVA accessibility depends entirely on cryptographic keys
- Key deletion → immediate model "forgetting" of user data
- Cryptographic proof of erasure through key destruction verification
- No residual information in base model weights

Detailed Technical Specifications

Cryptographic Implementation

Key Generation and Management:

```
Master Key: K_master ← CSPRNG(256 bits)
User Key: K_user ← HKDF-SHA256(K_master, user_id, "WDVA_v1")
Encryption: WDVA_blob ← XChaCha20-Poly1305.Encrypt(K_user, ΔW_user, AAD)
```

Associated Authenticated Data (AAD) Structure:

```
{
  "version": "WDVA_v1.0",
  "user_id": "sha256(user_identifier)",
  "created_timestamp": "2025-10-13T18:49:00Z",
  "data_sources": ["genomics", "wearables", "preferences"],
  "consent_scope": "fitness_coaching:90_days",
  "privacy_budget": {"epsilon": 0.5, "delta": 1e-5}
}
```

TEE Integration Specifications

Hardware Requirements:

- Intel SGX with at least 128MB EPC memory
- ARM TrustZone with secure world implementation
- AMD SEV-SNP with memory encryption support

Security Properties:

- Remote attestation for runtime verification
- Memory encryption with hardware-backed keys
- Secure interrupt handling during adapter merge
- Side-channel attack resistance through constant-time operations

Performance Characteristics

Benchmarked Performance Targets:

- Adapter decryption: < 5ms for 100MB weight deltas
- Ephemeral merge: < 25ms for 7B parameter models
- Memory footprint increase: < 10% over base model
- Revocation latency: < 2 seconds from key destruction to effect

WDVA Applications and Use Cases

Personalized Health and Wellness Ecosystem

Core Innovation: WDVA enables the creation of **personalized, continuously fine-tuned Small Language Models** that serve as secure vaults of personal data for fitness, genetic, and health wellness purposes, acting as sources of truth for agentic applications.

Revolutionary Application Architecture

Personal Health Agents as Trusted Intermediaries:
The WDVA system enables a new paradigm where agentic applications can provide highly personalized health and wellness guidance without ever accessing raw sensitive data. Through the MCP gateway, these agents operate under strict consent-based access controls.

Key Application Domains:

1. **Genomics-Informed Fitness Coaching**:
    - Integration of VCF files for personalized training recommendations
    - Nutrient metabolism optimization based on genetic variants
    - Recovery pattern adaptation using genetic predisposition data
    - All while maintaining strict non-diagnostic boundaries
2. **Continuous Biometric Integration**:
    - Real-time wearable data (HRV, sleep, VO_2) integration
    - Temporal pattern learning without persistent data storage
    - Adaptive training load recommendations
    - Circadian rhythm optimization based on personal patterns
3. **Holistic Wellness Orchestration**:
    - Cross-domain insights combining genetic, biometric, and behavioral data
    - Personalized supplement and nutrition recommendations
    - Stress management strategies based on individual response patterns
    - Sleep optimization protocols tailored to genetic and behavioral factors

Commercial Innovation:

- Users maintain complete ownership of their encrypted adapters

- True data portability across service providers
- Privacy-first monetization models without data extraction
- Regulatory compliance through technical measures rather than policy enforcement

Agentic Application Framework

Technical Innovation: The WDVA system enables agentic applications that can access personalized insights without ever seeing raw personal data through several novel mechanisms:

1. **Consent-Scoped Tool Access**: Agents can only access specific functionalities (e.g., "nutrition planning" vs. "clinical interpretation") based on dynamically granted consent tokens.
2. **Uncertainty-Aware Recommendations**: All outputs include calibrated confidence intervals, preventing overconfident claims in health-critical applications.
3. **Temporal Privacy**: Historical data influence without historical data retention through weight delta evolution.
4. **Audit Trail Maintenance**: Complete provenance tracking from raw data through feature extraction, adapter training, and final recommendations.

Prior Art Analysis and Novel Distinctions

Comparison with Existing Technologies

Traditional Federated Learning

Limitations:

- Requires data to leave user devices
- Vulnerable to gradient-based reconstruction attacks
- Cannot provide individual user revocation
- Lacks genomics-specific privacy protections

WDVA Advantages:

- Data never leaves user control through encrypted weight deltas
- Cryptographic revocation guarantees
- Genomics-aware uncertainty quantification
- Purpose-limited consent management

Existing Privacy-Preserving ML

Limitations:

- Differential privacy alone insufficient for genomics data
- No practical right-to-be-forgotten implementation
- Lacks healthcare-specific regulatory compliance frameworks
- No integration with modern consent management protocols

WDVA Innovations:

- Cryptographic deletion for immediate revocation
- Healthcare-regulation-compliant by design
- MCP integration for granular consent management

- Specialized genomics processing pipeline

Current Personalized AI Systems

Limitations:

- Centralized data storage creates privacy risks
- No effective way to handle sensitive genetic information
- Limited user control over data usage
- Vulnerable to data breaches and misuse

WDVA Breakthroughs:

- Decentralized personalization without central data repositories
- Genomics-first design with appropriate safeguards
- User-controlled, cryptographically-enforced consent
- Breach-resistant through encrypted weight delta architecture

Independent Patent Claims

Claim 1: Core Method

A method for privacy-preserving personalization of a small language model comprising:
(a) generating per-user weight-delta parameters from user's multimodal health data through parameter-efficient fine-tuning;
(b) encrypting said weight-delta parameters using authenticated encryption with associated data (AEAD) to create a sealed weight-delta vault adapter (WDVA);
(c) storing the WDVA with cryptographically-signed manifest containing consent scope and privacy budget information;
(d) issuing purpose-limited and time-scoped consent tokens via a Model Context Protocol (MCP) gateway;
(e) upon authorized request, ephemerally decrypting and merging said WDVA with a base small language model within a trusted execution environment to create a temporary personalized model instance;
(f) generating non-diagnostic recommendations with uncertainty quantification derived from genomics features; and
(g) implementing cryptographic right-to-be-forgotten through key destruction, rendering the WDVA permanently inaccessible.

Claim 2: System Architecture

A system comprising:
(i) a genomics feature extractor configured to process VCF files and output uncertainty-quantified genetic features;
(ii) a weight-delta vault storing encrypted adapters with provenance ledgers;
(iii) an MCP gateway implementing policy-based access control with hierarchical consent scopes;
(iv) a trusted execution environment runtime operative to perform ephemeral WDVA decryption and model merging;
(v) a consent management system issuing cryptographically-verifiable tokens with purpose and temporal limitations; and
(vi) a revocation system implementing immediate forgetting through cryptographic key destruction.

Claim 3: Computer-Readable Medium

A non-transitory computer-readable medium storing instructions that, when executed, cause a system to:

- encrypt per-user model adaptations into sealed vaults;
- verify purpose-limited consent tokens before model access;
- perform ephemeral weight-delta merging within isolated memory space;
- emit uncertainty-quantified recommendations for genomics-derived insights;

- maintain cryptographic audit trails linking data sources to outputs; and
- implement immediate revocation through cryptographic key destruction.

Dependent Claims and Technical Refinements

Advanced Security Features

1. **Zero-Knowledge Adapter Verification**: Cryptographic proofs that adapters were trained on authorized data without revealing training details.
2. **Homomorphic Computation**: Encrypted inference capabilities for ultra-sensitive genomics applications.
3. **Blockchain-Based Provenance**: Immutable audit trails using distributed ledger technology.

Performance Optimizations

1. **Adapter Compression**: Specialized compression algorithms for encrypted weight deltas.
2. **Batch Processing**: Efficient handling of multiple user requests with shared base models.
3. **Caching Strategies**: Secure caching of frequently accessed adapter components.

Healthcare-Specific Enhancements

1. **Clinical Decision Support Integration**: Interfaces with existing healthcare systems while maintaining privacy guarantees.
2. **Regulatory Compliance Automation**: Built-in compliance verification for healthcare regulations across jurisdictions.
3. **Family Privacy Protection**: Specialized handling of genomic data correlations between related individuals.

Technical Enablement and Implementation

Cryptographic Specifications

- **Encryption Algorithm**: XChaCha20-Poly1305 with 256-bit keys
- **Key Derivation**: HKDF-SHA256 with domain separation
- **Random Number Generation**: Hardware-backed CSPRNG
- **Memory Protection**: Constant-time operations with secure erasure

Performance Benchmarks

- **Encryption Throughput**: > 4 GB/s on modern hardware
- **Decryption Latency**: < 5ms for typical adapter sizes
- **Memory Overhead**: < 10% increase over base model inference
- **Revocation Time**: < 2 seconds for complete key destruction

Integration Requirements

- **TEE Support**: Intel SGX, ARM TrustZone, or AMD SEV-SNP
- **MCP Compatibility**: Full Model Context Protocol v1.0 support
- **OAuth Integration**: OAuth 2.0 with custom healthcare scopes
- **Audit Logging**: Immutable provenance tracking with hash chains

Regulatory Compliance and Safety

GDPR Article 9 Compliance

- Technical implementation of special category data protections
- Cryptographic enforcement of consent withdrawal
- Cross-border transfer safeguards through encryption
- Data subject rights implementation through technical measures

Healthcare Regulation Alignment

- FDA AI/ML SaMD guidance compliance for non-diagnostic systems
- HIPAA technical safeguards through cryptographic protection
- Clinical decision support boundaries with uncertainty quantification
- Medical device classification avoidance through design constraints

International Standards Compliance

- ISO/IEC 27001 information security management alignment
- NIST Cybersecurity Framework implementation
- Healthcare interoperability standards (FHIR) compatibility
- Privacy engineering best practices integration

Commercial Applications and Market Potential

Target Markets

1. **Personal Health Technology**: Wearable device manufacturers, fitness platforms
2. **Genomics Services**: Direct-to-consumer genetic testing companies
3. **Healthcare AI**: Clinical decision support system providers
4. **Enterprise Wellness**: Corporate health and wellness platforms
5. **Research Institutions**: Academic and pharmaceutical research organizations

Competitive Advantages

- **First-mover advantage** in privacy-preserving personalized health AI
- **Regulatory compliance by design** reducing compliance costs
- **User data sovereignty** enabling new business models
- **Scalable privacy architecture** supporting enterprise deployment
- **Healthcare-specific design** addressing unique sector requirements

Conclusion and Patent Scope

The Weight-Delta Vault Adapters (WDVA) invention represents a fundamental advancement in privacy-preserving AI personalization, specifically addressing the critical challenges of genomics data privacy, user consent management, and regulatory compliance in healthcare applications. The novel combination of encrypted weight deltas, ephemeral runtime merging, and MCP-gated consent management creates a comprehensive solution that enables unprecedented personalization without compromising individual privacy rights.

The invention's technical contributions include:

- **Novel cryptographic architecture** for AI model personalization
- **First practical implementation** of right-to-be-forgotten for neural networks
- **Healthcare-specific privacy engineering** for genomics data
- **Innovative consent management framework** using modern protocols
- **Scalable enterprise architecture** for privacy-preserving AI deployment

This patent proposal establishes comprehensive intellectual property protection for the WDVA technology stack, positioning it as a foundational technology for the next generation of privacy-first personalized AI systems in healthcare and wellness applications.

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